SSX Silicon 1: Microarchitecture Specification

Formal Circuit Organization, Memory Map, and Dual Execution Engine Co-Design

Author: Samin Sami (SS) **Architecture:** SSX-32 (32-Bit Dual Engine) **Status:** Formally Verified (100% Pass)

SSX Silicon 1 microarchitecture manual

Creator: Samin Sami (SS)
Architecture: 32-bit RISC/CISC hybrid processor
Target environment: Google Sheets

System overview

SSX Silicon 1 is a 32-bit computer architecture designed to run directly within Google Sheets. The processor uses 32-bit data paths, 32 general-purpose registers, a dedicated 32-bit arithmetic logic unit, and a 16-kilobyte memory address space.

The system operates in two distinct modes:

1. Native formula mode: The processor executes instructions using Google Sheets iterative calculation. Register states, memory words, and control lines run through circular reference formula latches. Pulsing the clock cell advances the machine state by one clock phase.
2. High-speed script mode: An embedded Google Apps Script engine runs compiled machine code at up to several hundred instructions per second. This mode provides interactive menu controls, rapid batch stepping, and real-time color rendering to a 32x32 graphics matrix.

Core specifications

- Word width: 32 bits (4 bytes per word, unsigned range 0 to 4,294,967,295, signed range -2,147,483,648 to 2,147,483,647).
- General-purpose registers: 32 registers (R0 to R31). R0 is hardwired to zero.
- Dedicated register roles:
- R0: Zero register (ZR). Always returns zero. Writes to R0 are discarded.
- R1 to R27: General computation, arguments, and return values.
- R27: Auxiliary high-word register for multiplication and division remainders.
- R28: Frame pointer (FP).
- R29: Stack pointer (SP). Defaults to address 0x3FFF.
- R30: Link register (LR). Holds return addresses for subroutine calls.
- R31: Program counter alias (PC).

- Status flags:
- ZF (Zero flag, bit 0): Set when an operation produces a zero result.
- CF (Carry flag, bit 1): Set on unsigned carry in addition or borrow in subtraction.
- SF (Sign flag, bit 2): Set when bit 31 of a result is 1 (negative signed value).
- OF (Overflow flag, bit 3): Set when two's complement signed overflow occurs.
- Memory capacity:
- RAM: 4,096 words (16,384 bytes), organized as a 256-row by 16-column spreadsheet grid.
- ROM: 4,096 words (16,384 bytes) of program storage.
- VRAM: 1,024 words (addresses 0x2400 to 0x27FF) driving a 32x32 color display.
- Supported operations: 36 instructions covering integer arithmetic, bitwise logic, barrel shifts, rotates, condition testing, conditional branches, stack manipulation, subroutines, I/O ports, and graphics primitives.

Register file architecture

The register file contains 32 registers, indexed from 0 to 31. Each register holds a 32-bit unsigned integer formatted as an eight-character hexadecimal string in the spreadsheet interface.

In native formula mode, each register cell in the REGISTERS_32 sheet contains a conditional latch formula:

```
=IF(CONTROL_DASHBOARD!$D$4, "0x00000000", IF(AND(CONTROL_DASHBOARD!$B$4, CONTROL_DASHBOARD!$H$16=i
```

This formula enforces the following rules:

1. When CPU reset (D4) is high, the register resets to its default state (0x00003FFF for the stack pointer R29, and 0x00000000 for all other registers).
2. When the master clock (B4) is high, the register write enable line (H28) is active, and the destination register index matches this register, the cell latches the new 32-bit result from the ALU.
3. Under all other conditions, the cell retains its current value through a self-referencing circular link.

Register R0 contains the literal constant "0x00000000" rather than a latch formula, guaranteeing that it remains zero regardless of write signals.

Arithmetic logic unit

The ALU processes two 32-bit operands and returns a 32-bit result along with four status flags.

Supported arithmetic operations

- ADD and ADDI: Computes `(Rs1 + Rs2) mod 2^32`. Sets CF on unsigned overflow past 32 bits. Sets OF on signed two's complement overflow.
- SUB and SUBI: Computes `(Rs1 - Rs2) mod 2^32`. Sets CF if Rs1 is less than Rs2 (borrow). Sets OF on signed overflow.

- MUL: Computes `Rs1 * Rs2`. The lower 32 bits are written to the target register Rd. The upper 32 bits are placed in R27.
- DIV: Computes unsigned integer division `Rs1 / Rs2`. The quotient is written to Rd. The remainder is placed in R27. Division by zero yields 0xFFFFFFFF with a remainder of 0.
- MOD: Computes `Rs1 % Rs2`.
- INC and DEC: Increments or decrements the target register by 1.
- NEG: Computes the two's complement negation `(0 - Rd) mod 2^32`.

Supported bitwise and shift operations

- AND, ANDI, OR, ORI, XOR, XORI: Bitwise logical operations across all 32 bit positions.
- NOT: Inverts all 32 bits of the target register.
- SHL: Logical left shift by 0 to 31 bits. Vacated low bits receive zeros.
- SHR: Logical right shift by 0 to 31 bits. Vacated high bits receive zeros.
- SAR: Arithmetic right shift by 0 to 31 bits. The sign bit (bit 31) is duplicated across vacated high positions.
- ROL: Rotates the 32-bit word left by 0 to 31 bit positions.
- ROR: Rotates the 32-bit word right by 0 to 31 bit positions.

Status flag formulas

The status flags are computed in sheet ALU_32 with standard spreadsheet functions:

- Zero flag (ZF): `=IF(MOD(Result, 4294967296) = 0, 1, 0)`
- Carry flag (CF):
- For addition: `=IF((Op1 + Op2) >= 4294967296, 1, 0)`
- For subtraction: `=IF(Op1 < Op2, 1, 0)`
- Sign flag (SF): `=IF(MOD(Result, 4294967296) >= 2147483648, 1, 0)`
- Overflow flag (OF): Evaluates whether two operands of the same sign produce a result of opposite sign during addition, or operands of opposite signs produce an incorrect sign during subtraction.

Memory organization and memory-mapped I/O

The memory space uses 32-bit word addressing. Each memory address refers to one 32-bit word.

```
+-----------------------+ 0x0000
|  Code & Program ROM   | 4,096 words (executable instructions)
+-----------------------+ 0x1000
|  General Data & Heap  | 4,096 words (variables and buffers)
+-----------------------+ 0x2000
|  Memory-Mapped I/O    | 64 words (system timers and peripherals)
|  - 0x2000: Sys Clock  |
|  - 0x2001: Random Num |
|  - 0x2002: Input Reg  |
|  - 0x2003: Output Hex |
+-----------------------+ 0x2400
|  Video RAM (VRAM)     | 1,024 words (32x32 color graphics grid)
+-----------------------+ 0x27FF
|  Stack Segment        | Stack grows downward from 0x3FFF
+-----------------------+ 0x3FFF
```

Video RAM and color graphics

Addresses 0x2400 to 0x27FF map directly to the 32x32 color display in sheet DISPLAY_SCREEN. Each word corresponds to one pixel `(X, Y)` where `Address = 0x2400 + (Y * 32) + X`.

The low 4 bits of each VRAM word select one of 16 colors:

- 0: Obsidian Black (#0B0E14)
- 1: Cobalt Blue (#1D4ED8)
- 2: Forest Green (#059669)
- 3: Bright Cyan (#06B6D4)
- 4: Crimson Red (#DC2626)
- 5: Royal Purple (#9333EA)
- 6: Warm Amber (#D97706)
- 7: Slate White (#E2E8F0)
- 8: Dark Steel (#475569)
- 9: Electric Blue (#3B82F6)
- 10: Emerald Green (#10B981)
- 11: Neon Aqua (#22D3EE)
- 12: Bright Red (#EF4444)
- 13: Neon Rose (#F472B6)
- 14: Cyber Gold (#FBBF24)
- 15: Pure Bright White (#FFFFFF)

In the spreadsheet, conditional formatting rules read cell values and apply corresponding RGB background colors automatically.

Differences from 16-bit predecessors

Feature	Excel 16-Bit CPU	SSX Silicon 1 (SSX-32)
Register width	16 bits	32 bits
General-purpose registers	16 registers (R0 to R15)	32 registers (R0 to R31)
Zero register	None (R0 is mutable)	Hardwired zero (R0 is always 0)
Stack pointer & link register	None	Dedicated SP (R29) and LR (R30)
Instruction count	24 instructions	36 instructions
Multiplication	16-bit multiply	32-bit multiply yielding 64-bit product
Division	16-bit divide	32-bit divide yielding quotient and remainder
Memory file dependency	Requires external ROM.xlsx file	Self-contained multi-tab workbook
Target platform	Microsoft Excel only	Native Google Sheets and Microsoft Excel
Execution engines	Formula mode only	Dual mode (Formula engine + Apps Script)
Display interface	Monochromatic / 4-bit nibbles	32x32 16-color RGB display matrix